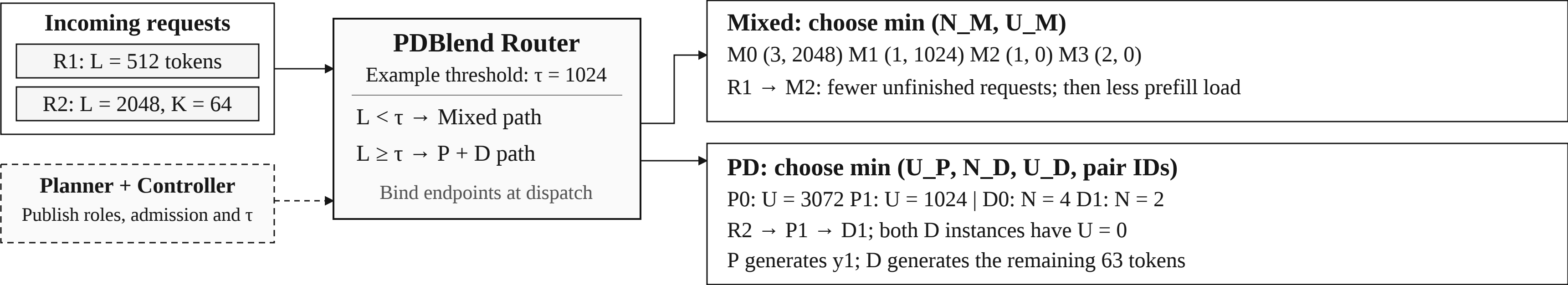


PDBlend: request routing and per-instance scheduling

■ Prefill ■ Decode ■ Wait / KV transfer ■ Output token

Default single-pool path · TP1 / PP1 illustration · vLLM V1

(a) Global routing: length threshold, then lexicographic instance load



N = unfinished sequences; U = prompt tokens awaiting first output. Counts include engine waiting and running work.
Only accepting instances; PD endpoints must be compatible. If a path is unavailable, use the other; if neither is available, reject.

(b) Inside each engine: continuous batching; FCFS applies to the waiting queue



Launcher defaults: 8192 scheduled tokens per iteration; 256 running requests; chunked prefill enabled.
This is not request-by-request execution to completion. KV pressure may cause preemption and recomputation.

(c) Example execution: separate P/D engines overlap; a Mixed engine can co-batch P and D

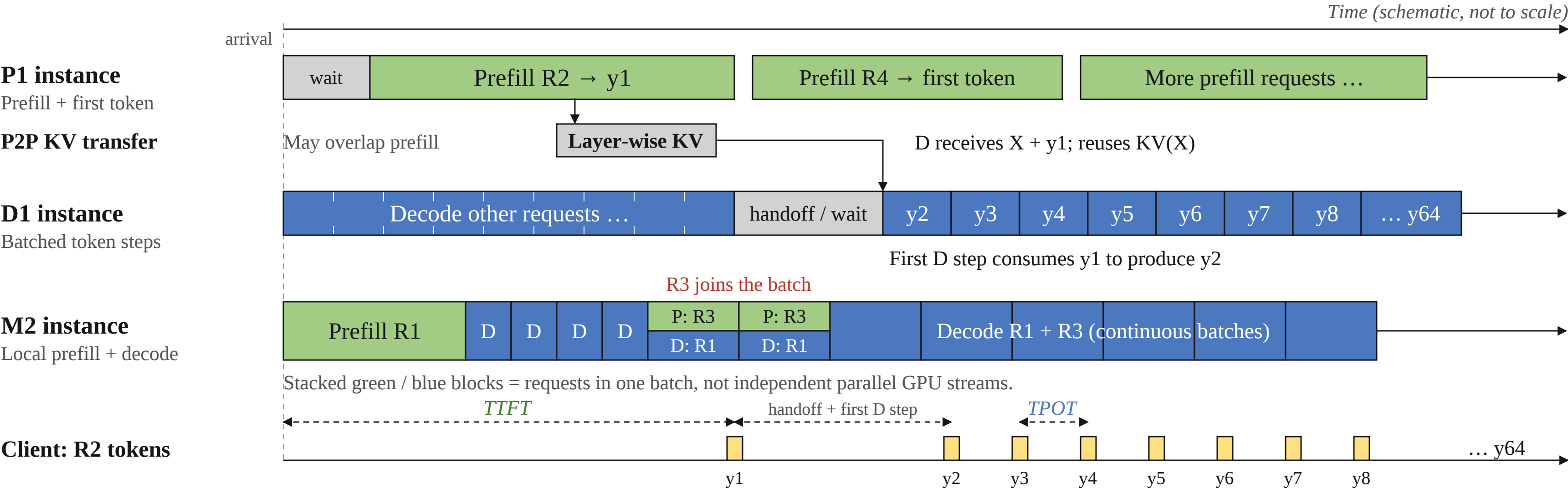


Figure. Default PDBlend scheduling. Panel (a) is a load snapshot; panel (c) is a separate illustrative schedule, not a measured trace.
Stable roles shown. Optional resident-pool / energy routing is outside this figure. FCFS does not imply global completion order.